

M M Mostafijur Rahman

Software Engineer

Phone/WhatsApp: +880 1877-344380 | Email: mustafiz.emon194@gmail.com | Tongi, Gazipur, Bangladesh
LinkedIn: linkedin.com/in/mostafijemon00 | GitHub: github.com/rahmanmostafijur
Portfolio: mostafij.vercel.app



PROFESSIONAL SUMMARY

Full Stack Engineer with a BSc in Computer Science and Engineering, working remotely for Acciptra (Japan) as the engineer building a multi-role e-commerce marketplace. Builds REST APIs in Python with FastAPI, PostgreSQL, SQLAlchemy and Alembic, and typed front ends in React, Next.js and TypeScript. Practical experience with JWT authentication, role-based access control, WebSockets, automated testing with pytest, and Docker-based delivery. Currently pursuing an MSc in CSE (Data Science) at East West University, with project work in RAG and Text-to-SQL systems.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Backend: FastAPI, Pydantic v2, SQLAlchemy 2 (sync and async), Alembic, REST APIs, WebSockets, JWT

Frontend: React, Next.js, Vite, Tailwind CSS, TanStack Query, React Router, React Hook Form, Zod, HTML, CSS

Databases: PostgreSQL, pgvector, PostgreSQL full-text search, schema design and migrations

Testing: pytest, Vitest, React Testing Library, Playwright

DevOps and Tools: Docker, Docker Compose, Nginx, GitHub Actions, Git, GitHub, Render, Vercel, Postman, Linux

AI and Data: RAG, Text-to-SQL, LLM API integration, embeddings and hybrid search, Pandas, TensorFlow/Keras, scikit-learn

Engineering: API design, authentication and authorization, RBAC, relational data modelling, state machines, audit logging, OOP, data structures and algorithms

PROFESSIONAL EXPERIENCE

Acciptra (Japan-based software company), Remote

Feb 2026 – Present

Software Engineer

Project: JapanHands — E-commerce marketplace for Japanese products (pre-launch)

FastAPI, PostgreSQL, React, TypeScript, Vite, Docker Compose, Postman

- Design and build the platform as the sole engineer: a buyer storefront, a seller dashboard and an admin panel, each a separate Vite + TypeScript app on its own subdomain behind a shared reverse proxy.
- Develop the Python backend with FastAPI and PostgreSQL, including role-based authentication and authorization for customer, seller and admin accounts.
- Built the product catalog with variants, search and filtering, and the cart, checkout and order workflow.
- Delivered seller and admin dashboards on top of the shared API, using React and TypeScript.
- Containerised the stack with Docker Compose, with self-hosted object storage for product media; test and document endpoints with Postman.

Pretty Group, S. Suhi Industrial Park Ltd. (Sweater Manufacturing), Ashulia, Savar

May 2024 – Jan 2026

Assistant Merchandiser

- Handled shipment work for export orders, coordinating between international buyers and factory teams.
- Operated buyers' technical portals and websites, translating buyer data and requirements into clear instructions for production.

SELECTED PROJECTS

Task Manager — Multi-user productivity platform

Live

FastAPI, PostgreSQL, SQLAlchemy, Alembic, pytest, Docker, GitHub Actions, React, TypeScript, Tailwind CSS

App: taskmanagerforyou.vercel.app | Code: github.com/rahmanmostafijur/taskmanager

- Built a REST API with JWT authentication using single-use, rotating refresh tokens that are revoked on password change, plus per-user data isolation.
- Implemented tasks, subtasks, projects, comments, an event-driven activity timeline, file attachments, time tracking, recurring tasks and notifications.

- Built a React + TypeScript client with list, drag-and-drop Kanban and calendar views, a command palette and a responsive layout down to 375px.
- Covered the backend with 126 pytest tests (96%+ line coverage); deployed the API on Render and the client on Vercel with GitHub Actions CI.

SupportDesk — Real-time customer support platform

FastAPI, async SQLAlchemy, PostgreSQL 16, WebSockets, React 19, TypeScript, Docker Compose, Nginx, Playwright

Code: github.com/rahmanmostafijur/SupportDesk

- Built live ticket chat over authenticated WebSockets with typing indicators, read receipts, presence, reconnect with missed-message catch-up and staff-only internal notes.
- Enforced four-role RBAC (customer, agent, manager, admin) on every request and socket frame; out-of-scope tickets return 404 so their existence is not revealed.
- Published real-time events only after the database transaction commits, so clients never receive rolled-back data.
- Added a knowledge base using PostgreSQL full-text search with weighted ranking; tested with 102 pytest, 41 Vitest and 8 Playwright end-to-end tests.

ProcuraFlow — B2B procurement and inventory platform

FastAPI, async SQLAlchemy, Pydantic v2, PostgreSQL 16, Alembic, React, TypeScript, Docker Compose

Code: github.com/rahmanmostafijur/ProcuraFlow

- Implemented a server-enforced purchase order state machine (draft to received, with cancellation) through a guarded transition table in the service layer.
- Designed an append-only inventory ledger: stock levels change only together with a ledger entry in the same database transaction.
- Built data-driven RBAC for five roles using permission tables and a reusable FastAPI dependency, with a full audit log of every mutating action.
- Delivered supplier performance, delayed-delivery and on-time-rate reports with CSV export.

NexaAI — Multilingual business AI agent (English, Bengali, Banglish)

FastAPI, PostgreSQL, pgvector, SQLAlchemy, sqlglot, React 19, TypeScript, Docker Compose

Code: github.com/rahmanmostafijur/NexaAI

- Built an agent that routes each question to read-only Text-to-SQL, hybrid RAG over company documents, or both, and streams cited answers over Server-Sent Events.
- Secured Text-to-SQL with AST validation (sqlglot), a read-only database role, statement timeouts and bounded self-correction.
- Implemented hybrid retrieval combining pgvector HNSW and PostgreSQL full-text search with Reciprocal Rank Fusion.
- Evaluated on an 80-question dataset: 98.8% routing accuracy and 82.1% SQL execution accuracy; backend covered by 211 tests.

Other projects on GitHub: Solmira (multi-tenant order and inventory platform, Next.js and NestJS, in progress); Namao (FastAPI + yt-dlp video downloader packaged as a Windows executable).

EDUCATION

MSc in Computer Science and Engineering (Major: Data Science)

Jun 2026 – Present (expected 2027)

East West University, Dhaka

BSc in Computer Science and Engineering

2019 – 2023

Green University of Bangladesh, Dhaka

- Final-year project: IoT and machine learning air quality monitoring system (Arduino sensors streaming to ThingSpeak); a Keras MLP reached 0.89 accuracy against a 0.66 baseline.

HSC, Science — GPA 5.00/5.00

2017

Adamjee Cantonment College, Dhaka Board

SSC, Science — GPA 5.00/5.00

2015

Moulovi Abdul Hye Memorial High School, Dhaka Board

ADDITIONAL INFORMATION

Languages: Bangla (native), English (fluent)

Work preference: Remote or on-site in Dhaka